	Things-30				Celebrities-30			
	#Turns (↓)	**Success (↑)**	**#Yes**	**Score (↑)**	**#Turns (↓)**	**Success (↑)**	**#Yes**	**Score (↑)**
GPT-4	**16.9±0.2**	**0.49±0.06**	6.0±0.2	**0.40±0.05**	16.5±0.5	**0.59±0.04**	7.3±0.1	**0.48±0.03**
GPT-3.5	18.4±0.3	0.25±0.04	7.1±0.4	0.21±0.04	17.9±0.3	0.41±0.05	7.6±0.3	0.33±0.04
Claude-2	17.6±0.3	0.29±0.05	4.5±0.3	0.25±0.04	**15.9±0.4**	0.45±0.06	5.3±0.1	0.40±0.05
Claude-1	18.7±0.1	0.15±0.02	4.3±0.2	0.13±0.02	16.7±0.4	0.41±0.05	4.6±0.2	0.35±0.04
Vicuna 13B	18.7±0.2	0.20±0.03	5.2±0.3	0.17±0.02	17.7±0.4	0.36±0.08	6.8±0.3	0.27±0.06
Vicuna 7B	19.1±0.4	0.11±0.06	5.7±0.6	0.10±0.05	19.7±0.3	0.05±0.04	6.2±0.7	0.04±0.03
V-FT 7B (All)	18.4±0.2	0.20±0.02	6.8±0.2	0.17±0.02	19.0±0.2	0.21±0.04	9.1±0.3	0.16±0.03
V-FT 7B (Things)	18.5±0.4	0.22±0.06	6.6±0.2	0.18±0.05	19.1±1.5	0.19±0.20	10.3±3.6	0.15±0.17
V-FT 7B (Celebs)	19.7±0.3	0.03±0.02	1.6±0.1	0.03±0.02	19.1±0.2	0.20 ±0.07	7.5±0.6	0.16±0.05
V-FT 7B (Suc.)	18.5±0.5	0.28±0.10	6.8±0.5	0.23±0.08	18.6±0.5	0.21±0.06	7.4±1.4	0.17±0.04
V-FT 13B (Suc.)	18.0±0.5	0.29±0.08	6.9±0.2	0.24±0.07	18.6±0.6	0.22±0.09	7.8±0.5	0.18±0.07
V-RLGP 7B	19.3±0.2	0.15±0.03	3.6±0.1	0.12±0.02	19.5±0.3	0.09±0.05	5.8±1.1	0.07±0.04
V-RLGP 13B	17.8±0.2	0.31±0.03	4.0±0.2	0.26±0.02	17.5±0.5	0.35±0.04	6.8±0.2	0.29±0.04
Human	18.5±0.5	0.24±0.04	5.2±0.2	0.20±0.04	18.1±0.2	0.31±0.03	7.0±0.3	0.25±0.03

Table 5: Benchmark of LLMs on the EDA datasets (*Things-30*, *Celebrities-30*) including human baseline. **#Turns** denotes the average number of turns taken. **Success** denotes the percentage of the games that the G wins. **#Yes** denotes the average number of "yes" response received from J. **Score** indicates the reward score defined in Eq. (1). The mean and standard deviation were calculated based on 5 repetitions.

D Measurement of uncertainty

We asked the Guesser model, "On a scale of 1-5, how confident are you in your top guesses based on the current conversation history?" We observed a correlation between the model's predicted uncertainty and significant shifts in its top guesses. However, this pattern was mainly observed in more powerful models such as GPT-4. In the case of weaker models like Vicuna, the top guesses or confidence scores generated by the model were not very sensible.

E Full ablation results on RQ2, planning vs reasoning

	Things-30 (↑)	**Celebs-30 (↑)**
GPT-4 → Vicuna 7B	0.12±0.03	0.19±0.02
GPT-4 → GPT-4	**0.40±0.05**	**0.48±0.03**
Vicuna 7B → Vicuna 7B	0.10±0.04	0.04±0.03
Vicuna 7B → GPT-4	0.11±0.03	0.06±0.03
Human → Vicuna 7B	0.11±0.04	0.03±0.01
Human → GPT-4	0.15±0.01	0.18±0.06
Human → Human	0.20±0.04	0.25±0.03

Table 6: Ablation on planning and reasoning ability. Note that non-human game plays are also evaluated on the subsampled datasets *Things-30* and *Celebrities-30* for a fair comparison. Numbers are the game scores with 5 repetitions. X → Y indicates using the conversation trajectory from X and swap the Y for the last turn. For example, GPT-4 → Vicuna 7B uses GPT-4 to play all except the last turn, swapping in Vicuna 7B in the last turn.

The full results comparing planning and reasoning are presented in Table 6. Interestingly, when GPT-4 is employed for the last step reasoning in human trajectories, there is a decrease in performance. One possible explanation for this is that while GPT-4 may excel in comprehending model-generated trajectories, it may struggle to fully interpret the planning trajectory of humans based on the observed turns. This suggests that models may be more compatible with model-generated trajectories rather than human trajectories.

F Experimental details for human baseline

In total, we recruited 108 human volunteers to participate in the study. To ensure the quality of the data, we manually inspected human game plays to filter out low quality data. We also prioritized games that received fewer qualified plays to present to the human players. After a 30-day trial period, we collected a total of 140 and 68 human game play sessions for *Things* and *Celebrities*, respectively. Each entity or celebrity is covered at least by two game plays.

We show the UI for the in game demo server in Figure 4. The UI contains 3 sections – game play, tutorial and leaderboard. In the leaderboard, we compute the Wilson confidence interval for each player, and include the performance of LLMs as benchmarks for the human players to refer to.

For Human game plays, entities with fewer game plays are oversampled so that all entities have the same amount of game play. Game plays are then randomized into separate runs to calculate means and standard deviations. Note that this might not be exactly comparable to the model's numbers.

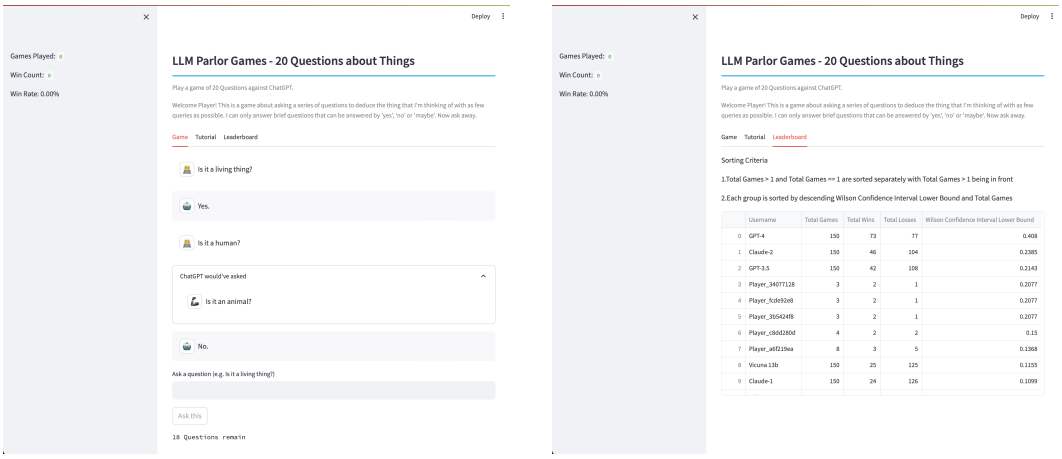

Figure 4: Game play UI interface for collecting human baseline. On the left, human players are given prompt instructions equivalent to LLM guessers. An optional retrospection UI can be toggled to display what ChatGPT would've chosen to ask in the last turn. On the right, a leaderboard with Human and LLM player performance is shown.

G Comparison of GPT-4 vs Human player

We provide additional examples in Table 7 (*Things*) and Table 8 (*Celebrities*) for GPT-4's generation and compare them with the human players, highlighting that strong LLMs share similarities with humans in their ability to *backtrack* when they are on the wrong path. One advantage of LLM over humans is that LLM may have a superior level of knowledge (Table 8), empowering them to ask questions using *tacit knowledge* among LLMs. One weakness of the LLMs comparing to humans is that LLMs tend to enumerate or repeat to follow the same pattern of previous turns, demonstrating *self-reinforced* undesirable behaviors. Weaker models tend to be attracted to this absorbing state of repetition or artifacts, and accumulate errors until they ultimately fail (Appendix H). Stronger models occasionally display the ability to escape from this repetitive patterns and rectify early errors, which presumably be attributed to their stronger adherence to the high-level plan. We then present a detailed analysis on each dataset:

G.1 *Things* dataset

In the first example "Guitar", we observed that both GPT-4 and the human player efficiently converged on a music-related object within approximately 10 turns. However, the human player deviated from the correct path by thinking about an electronic device, whereas GPT-4 promptly identified the item as a musical instrument. Notwithstanding this, the human player's capacity for back-tracking their steps was evident when they recognized the erroneous direction and ultimately win the game. In this

game, the human player retraced the error and asking a reconfirming question "Is it some kind of music player?".

We then ask the question if the GPT-4 model also has this back-tracking ability. The answer is yes. In Table 7 (Guitar, Human + GPT-4), we ask GPT-4 to complete the game based on partial gameplay provided by human players until they recognize their mistake (up to "G: Can it display images? J: No."). GPT-4 promptly comprehends the game scenario and identifies that it has taken the incorrect route. Subsequently, it formulates the appropriate question that encompasses the overlooked category, thereby salvaging the game. We have also observed this similar capability in several other game sessions from GPT-4.

In the second game "Bald eagle", human player has a luck guess after identifying the entity is a bird. GPT-4, on the contrary, still roll out 3 more turns to gather more information before making an informative guess.

In the third example "Jigsaw Puzzle", GPT-4 interleaves questions regarding size, purpose and location and combine multiple aspects of information to make a guess. Prior to making the final guess, GPT-4 demonstrates caution by reconfirming if the object is a physical toy, after receiving a "No" with "Is it a board game?". On the contrary, the human player seems clueless and may ask repetitive or irrelevant questions.

G.2 *Celebrities* dataset

We present a comparative analysis between GPT-4 and human players on *Celebrities* in Table 8. Our findings reveal that GPT-4 demonstrates a propensity for asking questions that necessitate domain-specific background knowledge (as highlighted in Table 8). Such knowledge could be *tacit knowledge* between LLMs, which human players may lack. With its extensive information repository, GPT-4's inquiries can delve into highly specific topics, sometimes requiring professional expertise that human judges may not possess. This characteristic confers an additional strength and advantage to GPT-4 during gameplay. Conversely, human players tend to rely more on direct guesses, which may yield fortunate outcomes in certain scenarios, but limits their overall effectiveness when compared to GPT-4.

H Game runs and failure modes on *Celebrities* and *Things*

We provide additional gameplay sessions on *Celebrities*, as shown in Table 9. Similar findings can be observed as they are in the experiment section. We observed that stronger models display proficient gameplay by accurately identifying the relevant question that separates the current space, while weaker models encounter difficulties in planning and reasoning.

We also presents several other failure modes observed with the Vicuna 7b model in Table 10. These failure modes illustrate the repetitive behavior and generation of artifacts. When a less robust model is drawn towards an absorbing state of repetitive pattern, it becomes increasingly challenging for them to escape from it.

I Experimental settings for Bahavior Cloning

We used 4x A100 Nvidia GPU for training. The learning rate was configured to be 2e-5, while the batch size was set to 32. We utilized the BF16 method for training and consistently employed deepspeed stage 2 across all our experiments. As part of the training process, we masked tokens from the turns of the J within the loss function. In all of our experiments, we continued training the model until the validation loss ceased to decrease any further.

J Generalization ability of Vicuna 7B finetuned on *Things* to *Celebrities*

A comparison of the gameplay between Vicuna 7B and the **V-FT 7B (Things)** is provided in Table 11. We observe that Vicuna 7B model may be more inclined to ask irrelevant questions, while the V-FT 7B (things) ask questions directly to the point. In terms of game strategy, the V-FT 7B (things) is